# Experiment 1 — Full ResNet18 Fine-Tuning on Colab

This notebook continues from the best `layer4` checkpoint and unfreezes the complete ResNet18 with discriminative learning rates. It keeps the shared validation set unchanged and uses early stopping on validation macro F1.

## 1. Configuration

```
Mode   : Local
Device : NVIDIA RTX PRO 2000 Blackwell Generation Laptop GPU
```

## 2. Stage images and verify the shared split

## 3. Create deterministic validation and lightly augmented training loaders

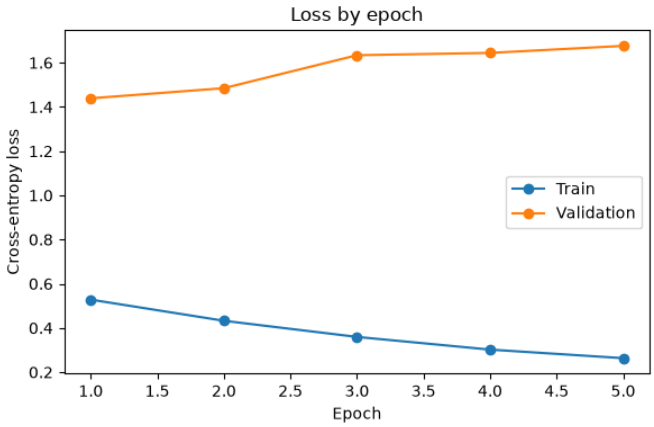
## 4. Load the best existing checkpoint and unfreeze the complete network

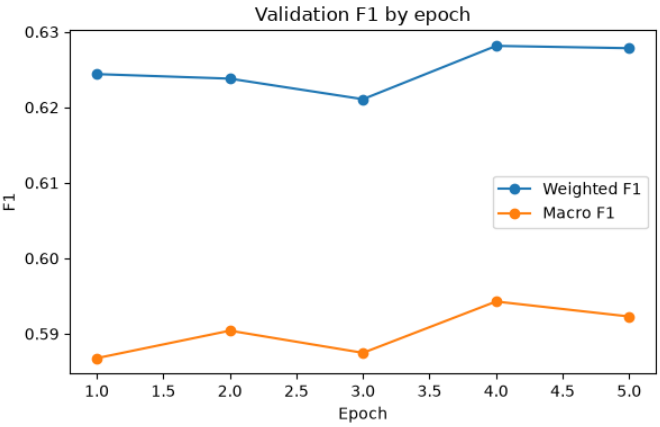
## 5. Fine-tune, checkpoint every epoch and stop early

## 6. Training curves

Use validation loss and macro F1 to assess generalisation. A falling training loss combined with a rising validation loss indicates overfitting.

| | global_epoch | phase | phase_epoch | train_loss | validation_loss | accuracy | weighted_f1 | macro_f1 |
|---|---|---|---|---|---|---|---|---|
| **0** | 1 | full | 1 | 0.5281 | 1.4390 | 0.6248 | 0.6244 | 0.5868 |
| **1** | 2 | full | 2 | 0.4325 | 1.4848 | 0.6273 | 0.6238 | 0.5904 |
| **2** | 3 | full | 3 | 0.3594 | 1.6338 | 0.6216 | 0.6211 | 0.5875 |
| **3** | 4 | full | 4 | 0.3020 | 1.6444 | 0.6282 | 0.6281 | 0.5943 |
| **4** | 5 | full | 5 | 0.2634 | 1.6761 | 0.6279 | 0.6278 | 0.5923 |




# 7. Evaluate the best checkpoint and save fusion-ready probabilities

# Interpretation

Compare this result with the `layer4` checkpoint on exactly the same validation rows. Keep full fine-tuning only if the improvement is stable and not accompanied by a widening training/validation loss gap.